

ACT-BP Series Drillable Cast Iron Bridge Plugs (4-1/2 and 5-1/2 in.)

Technical Datasheet DS-BP-SERIES Rev D | Product family: ACT-BP series

1. Product Overview

ACT-BP series bridge plugs are high-pressure drillable cast iron plugs for temporary or permanent zonal isolation. Run on electric wireline with the ACT-ST-10/ST-20 setting tools (adapter kits AK-45/AK-55), or on tubing or coiled tubing with a mechanical setting tool. Fully drillable construction mills out quickly with standard junk mills.

2. Applications

- Zone abandonment and temporary isolation
- Casing pressure tests
- Below-perforation isolation for stimulation
- Well suspension

3. Features

- All cast iron Class 40 construction - drillable in under 45 minutes
- Three-piece HNBR element with expandable soft-steel back-up rings
- Bi-directional segmented slips with hardened inserts (55 HRC)
- Sets on standard #10/#20 wireline pressure setting tools
- Tension-nut release with controlled shear-out
- Qualified to API 11D1 Grade V3 (liquid)

4. Technical Specifications

Parameter	ACT-BP-4500	ACT-BP-5500
Casing Size	4-1/2 in., 9.5-13.5 lb/ft	5-1/2 in., 14-23 lb/ft
Maximum OD	3.750 in.	4.437 in.
Overall Length	22.5 in.	24.8 in.
Pressure Rating (above/below)	10,000 psi	10,000 psi
Temperature Rating	300 degF	300 degF
Setting Tool / Kit	ACT-ST-10 + AK-45	ACT-ST-10 + AK-55 or ACT-ST-20
Tension Nut Shear	32,000 lbf +/- 10%	38,000 lbf +/- 10%
Slips (upper/lower)	4 / 4 segments	6 / 6 segments
Avg. Mill-Out Time (field)	35 min	42 min

All ratings are at ambient temperature unless otherwise noted. Performance envelope per API 11D1 V3 qualification testing where indicated. Consult ACT Engineering for combined load conditions.

5. Materials of Construction

Component	Material
Mandrel / cones / guide	Cast iron Class 40 (ASTM A48)
Slips	Cast iron CL 40 with heat-treated inserts, 55 HRC

Component	Material
Element	HNBR 80/90/80
Back-up rings	AISI 1008 soft steel
Tension nut	AISI 12L14

6. Ordering & Reference Documents

Order by part number and specify: casing size and weight, service trim (standard or NACE sour), elastomer option, end connections, and any customer inspection requirements. Related documents:

- BOM-BP-4500 Rev D
- BOM-BP-5500 Rev C
- MAN-BP Series Manual
- ECO-2025-091 (slip insert upgrade)

Datasheet DS-BP-SERIES Rev D issued 2025-09-30. Prepared by R. Callahan; approved by J. Martinez, P.E.. Specifications subject to change; confirm with ACT Engineering for well-specific designs.